

Question:

Janet’s ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for \$2 per fresh duck egg. How much in dollars does she make every day at the farmers' market?

Reasoning + Answer:

<think>

Janet eats 3 eggs / breakfast * 16 eggs / breakfast = 48 eggs .

She bakes 4 muffins * 16 eggs = 64 eggs .

Total eggs used is 48 eggs + 64 eggs = 112 eggs .

Total eggs remaining is 16 eggs - 112 eggs = - 96 eggs . This is not possible so we know that some of the eggs must have been used up in the muffins or at breakfast . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 16 breakfasts = 48 eggs .

She bakes 4 muffins * 12 eggs = 48 eggs .

Total eggs used is 48 eggs + 48 eggs = 96 eggs .

Total eggs remaining is 16 eggs - 96 eggs = - 80 eggs . This is not possible so some other eggs must have been used up . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 15 breakfasts = 45 eggs .

She bakes 4 muffins * 13 eggs = 52 eggs .

Total eggs used is 45 eggs + 52 eggs = 97 eggs .

Total eggs remaining is 16 eggs - 97 eggs = - 81 eggs . This is not possible so some other eggs must have been used up . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 14 breakfasts = 42 eggs .

She bakes 4 muffins * 14 eggs = 56 eggs .

Total eggs used is 42 eggs + 56 eggs = 98 eggs .

Total eggs remaining is 16 eggs - 98 eggs = - 82 eggs . This is not possible so some other eggs must have been used up . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 13 breakfasts = 39 eggs .

She bakes 4 muffins * 15 eggs = 60 eggs .

Total eggs used is 39 eggs + 60 eggs = 99 eggs .

Total eggs remaining is 16 eggs - 99 eggs = - 83 eggs . This is not possible so some other eggs must have been used up . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 12 breakfasts = 36 eggs .

She bakes 4 muffins * 16 eggs = 64 eggs .

Total eggs used is 36 eggs + 64 eggs = 100 eggs .

Total eggs remaining is 16 eggs - 100 eggs = - 84 eggs . This is not possible so some other eggs must have been used up . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 11 breakfasts = 33 eggs .

She bakes 4 muffins * 17 eggs = 68 eggs .

Total eggs used is 33 eggs + 68 eggs = 101 eggs .

Total eggs remaining is 16 eggs - 101 eggs = - 85 eggs . This is not possible so some other eggs must have been used up . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 10 breakfasts = 30 eggs .

She bakes 4 muffins * 18 eggs = 72 eggs .

Total eggs used is 30 eggs + 72 eggs = 102 eggs .

Total eggs remaining is 16 eggs - 102 eggs = - 86 eggs . This is not possible so some other eggs must have been used up . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 9 breakfasts = 27 eggs .

She bakes 4 muffins * 19 eggs = 76 eggs .

Total eggs used is 27 eggs + 76 eggs = 103 eggs .

Total eggs remaining is 16 eggs - 103 eggs = - 87 eggs . This is not possible so some other eggs must have been used up . Let’s recalculate that .

Janet uses 3 eggs / breakfast * 8 breakfasts = 24 eggs .

She bakes 4 muffins